Presentation Video — narration, v1

iGEM Claremont-CA 2026 | draft of 2026-10-24

Timecodes assume 140 words per minute; the uploaded file has a hard limit of 15:00.

Line numbers run continuously — send edits back as *line 212: …*

Cold open

126 words ends 0:54

ON SCREEN — Plaque assay animation; then the same drop on two other strains, no effect.

A drop of phage goes onto a lawn of bacteria. A few hours later, there's a clear circle where the bacteria used to be.

Move the same drop to the plate next door. Same phage. Different strain. Nothing.

Phages are the most precise killers we know of — and that precision is the problem. If you want to treat an infection with a phage, you need a phage that matches the patient's strain. Most of the time, there isn't one on the shelf.

What decides the match is the very first step. The phage has to recognise the cell surface and stick to it. And for bacteria that wrap themselves in sugar, that surface isn't protein at all.

So what is the phage's tail actually recognising?

Act I.1 | Our system

69 words ends 1:23

ON SCREEN — Sampling map (check: no flags), bacterial genomes, phage genome. Credit the phage assembly to the PI.

We went looking for our own system — water and soil from around our campus.

We came back with three strains of *Pseudomonas*, distinct from one another, and one phage, Phi-8, that infects exactly one of them. About sixty-two kilobases, sharing almost no genes with anything described before. The three bacterial genomes are ours — sequenced, assembled and published. The phage genome was assembled and annotated by our PI.

Act I.2 | The first clue

71 words ends 1:54

ON SCREEN — The three O-antigen loci side by side, labelled `prediction --- not yet tested`. This figure returns at the end of Act II.

Then we lined up the three strains at the one locus that builds their sugar coat.

Same position in the genome. Same gene on either side. And inside — three cassettes with no homology to each other at all. Of forty-five gene clusters, four were shared.

Three strains. Three different sugar coats. One phage that kills one of them.

The prediction is sitting right there. Whatever Phi-8 recognises, it's that sugar.

Act I.3 | The textbook

53 words ends 2:16

ON SCREEN — Lock and key. First appearance of the beta-helix tail spike render, labelled `the key?`

But we didn't follow the sugar. We followed the textbook.

The textbook says a phage's receptor-binding protein meets a protein on the bacterial surface. Lock and key — protein against protein. Work out which binding protein fits which receptor, and you've worked out host range.

So that's what we set out to do.

Act II.1 | Why a computer

125 words ends 3:10

ON SCREEN — 114 × 63 = 7,182 matrix; then an interface score bar from 0 to 1.

We had a hundred and fourteen candidate binding proteins, and sixty-three candidate receptors. That's seven thousand, one hundred and eighty-two combinations. Testing those at the bench is years of work.

So we did what the field does. We asked a computer.

Programs like AlphaFold predict the shape a protein folds into. Give them two proteins, and they'll also predict how the two fit together — and hand you back a number between zero and one. An interface score. High means: I'm confident these two dock, like this.

The plan was to score all seven thousand pairs, and let the top of that list tell us where to start in the lab.

Most of the next few months went into getting it to run at all.

Act II.2 | The controls

181 words ends 4:27

ON SCREEN — Control design, then the controls comparison figure. Hold 0.137 and 0.136 large on screen.

And then, late — too late — we added controls.

A control here is a simple idea. Before you trust a method on a question you can't check, you give it questions you can.

We picked three phage-receptor pairs that are known to bind. Known well enough that somebody has solved their structures experimentally, atom by atom. Then we built fakes — the same receptor with its sequence shuffled into nonsense, and pairs deliberately mismatched.

If the method can tell binding from non-binding, the real pairs have to beat the fakes. That's the only bar it has to clear.

The best-characterised pair we had was T-five's pb-five with FhuA — a complex that has been co-crystallised. It scored zero point one three seven.

The same receptor, with its sequence shuffled into gibberish, scored zero point one three six.

One thousandth apart.

We assumed the tool was at fault, and rebuilt the whole thing with a different program, using real evolutionary data. It got worse. A deliberately mismatched pair scored zero point five — higher than every true binder in the set.

Act II.3 | What was going wrong

316 words ends 6:43

ON SCREEN — Two confidence bars: on scrambling the left collapses 0.55 to 0.17, the right barely moves. Then paired-MSA diagram: two circles that never intersect.

One detail told us what was actually going on.

These programs report two separate confidences. One is about the shape of the molecule itself. The other is about how the two pieces fit together.

When we shuffled the sequence, the first number collapsed — from zero point five five down to zero point one seven. So the model noticed. It knew the shuffled protein wasn't a real protein.

But the second number — the fit — barely moved.

So the model wasn't being fooled by our fake. It had simply never had an opinion about the fit. Not for the fake, and not for the real pair either. It had been handing us a number that looked like an answer and wasn't one.

Here's why.

When AlphaFold predicts how two proteins touch, it isn't simulating physics. It's reading evolutionary history. It looks across thousands of species, finds both proteins in each one, and asks one question: when this spot on protein A changes, does that spot on protein B change to match? Two surfaces that have to keep fitting each other keep changing in step. That pattern is the fingerprint of contact.

But that record only exists if the two proteins have been inherited together — in the same genome, for millions of years.

A phage protein's relatives live in other phages. A bacterial protein's relatives live in other bacteria. They are never in the same genome. They

have never been inherited together.

So for a pair like ours, the evolutionary record of that interface isn't hard to find. It was never written.

Which means this was never a result about our proteins. It's a limit of the method — and it holds for any phage and host, and for host and pathogen across kingdoms generally. More compute doesn't fix it.

Ten control folds decided the fate of seven thousand. We never ran the real screen.

Act II.4 | The fold tells the truth

170 words ends 7:56

ON SCREEN — Beta-helix render returns, H-bond period marked, caption `CDS_0011 res 540--941`. Ends by replaying the Act I.2 figure.

So we stopped asking what the tail spike binds, and asked what it is. We folded it on its own.

It came back as a right-handed beta-helix — a long triangular prism with a groove running down the side.

That shape is a signature. Enzymes that cut long sugar chains look like this, and they look like this for a reason. A sugar chain is a long thread, and to cut it you need something that holds it straight and feeds it through. You can recognise a wood plane by its shape without being told what it cuts.

Our tail spike wasn't a key. It was a blade — an enzyme for sugar.

Which turns the whole question over. If the tail spike cuts sugar, then

what the phage recognises is the sugar coat, and the protein lock we'd spent months looking for doesn't exist.

The identity of the receptor didn't come from the docking. It came from the shape.

And it points straight back to those three sugar coats.

Act II.5 | The hypothesis forms

53 words ends 8:18

ON SCREEN — Attribution has to be exact here.

We weren't the only ones reading it that way. Our own RBP-detection model had already flagged the protein as a depolymerase — and at about the same time, our PI reached the same conclusion from his own structural search.

What he added was the step from an observation to something you can test.

Act III.1 | The hypothesis

83 words ends 8:54

ON SCREEN — One-sentence hypothesis box on white, held. Credited to the PI.

In bacteria that wrap themselves in sugar, the ceiling on a phage's host range is set by the substrate range of its depolymerase — not by the identity of its binding protein. And the narrow specificity we see today is a narrowed-down version of something broader in the past.

That changes what you'd test. Not binding. Substrate range.

And if today's enzymes are narrowed-down descendants, then the interesting experiment is to go back up the tree, and ask what the ancestor could do.

Act III.2 | The targets

57 words ends 9:18

ON SCREEN — Phage genome zoomed to the tail module. Canonical CDS numbering.

Phi-8's tail module has two enzymes in it. The tail spike we've just been talking about — CDS eleven, with that beta-helix at its business end. And next to it, CDS twelve, a glycoside hydrolase from family five. There's also CDS ten, which is still an unknown protein. We expressed it to find out what it is.

Act III.3 | Resurrecting the ancestor

126 words ends 10:12

ON SCREEN — Tree collapsing back to ancestors, the three nodes, identity matrix, perplexity trend. Phylogeny, ASR and scoring credited to the PI. RMSD values stay on screen, unspoken.

Our PI took CDS twelve, aligned its closest relatives, built a phylogeny, and reconstructed the ancestral sequence at every node on the path back to the root. Then he scored each reconstruction with a protein language model. The further back he went, the better the model liked it.

We picked three of those nodes and had them synthesised. One is seventy-eight percent identical to the modern enzyme. The other two are about half identical to it — but eighty-seven percent identical to each other. So the panel is the modern enzyme, one close relative, and a pair of more

distant cousins.

Before any bench work, we asked whether those resurrected sequences still fold into the same enzyme. All three come back as the same TIM barrel.

Act III.4 | The constructs

61 words ends 10:39

ON SCREEN — Six constructs. Put res 305–941 and res 540–941 on one axis so the overlap is visible.

Six constructs, all with an N-terminal histidine tag, synthesised by Twist. The beta-helix piece of the tail spike itself. CDS twelve. Its three ancestors. And the unknown protein.

That first one matters. The fragment we ordered contains exactly the beta-helix domain we'd identified by folding it. What the shape told us to look at is what we put in a tube.

Act III.5 | The bench

107 words ends 11:24

ON SCREEN — Gel with the target boxed in red. On screen: `IPTG -/+ : present | empty-vector control: RUNNING NOW | n = 1`

And then the bench — which is where we are right now.

When we induce CDS twelve, we see a band in the right size range. To separate that band from host background we need an empty-vector control, and that control is on the bench as I'm speaking.

The nickel purification of one ancestor failed. The elutions came off empty, and the flow-through looked like the load — so the protein never

bound the column. We're repeating it with the binding conditions changed.

So the enzyme arm is honest work in progress. I'd rather tell you exactly which part is unfinished than show you a cleaner story.

Act III.6 | One clean result, and safety

72 words ends 11:55

ON SCREEN — Read pileup with F126L marked; sugar locus `unchanged`; `0 VFDB / 0 CARD` and IBC approval.

One line did come back clean.

We raised mutants resistant to Phi-8, and sequenced them. One carries a single amino-acid change in a gene that builds the capsule — confirmed down at the level of the sequencing reads. And the sugar locus itself is untouched.

On safety: Phi-8's genome carries no virulence factors and no antibiotic-resistance genes. Zero hits in either database. And the work has approval from our institutional biosafety committee.

Coda | Where we are

132 words ends 12:52

ON SCREEN — Three columns fading in: Established / Predicted, not tested / Open.

So, plainly.

Established: the genomes. Three unrelated sugar cassettes in one slot. A tail spike that turns out to be a sugar-cutting enzyme. A controlled negative result about interface prediction. One resistance mutation, con-

firmed in the reads.

Predicted, but not yet tested: that Phi-8 won't infect the other two strains. That the ancestral enzymes have a broader substrate range.

Still open: clean expression, an activity assay, and the topology of the phage genome.

The cheapest next step is that control — and it's running. The decisive one is knocking out the sugar and watching whether the phage still sticks.

And if there's one thing we'd hand to anyone starting a screen like ours — gate it on controls before you trust it.

Ten folds told us the truth that seven thousand couldn't.

1802 spoken words. With three act cards and a 30-second credit card, the finished file runs about 13:30 at 140 words per minute.